

MATHEMATICAL LECTURE NOTES ON ANALYTICAL GEOMETRY

Polar Coordinates: Straight Line, Circle, Conic Sections, Tangents, Normals & Chords

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1 Module 1: Polar Coordinates and the Polar Straight Line

1.1 Polar Coordinate System

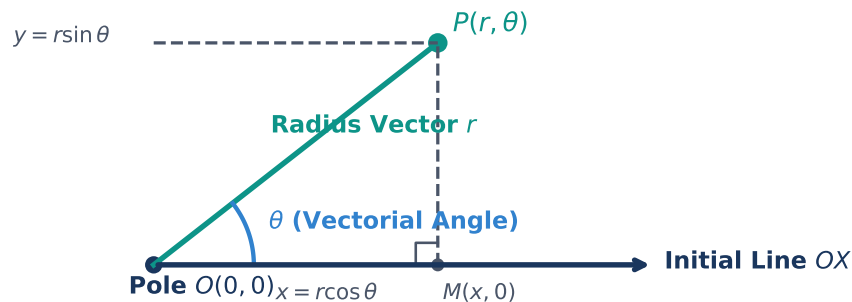
Definition: Polar Coordinates

Let O be a fixed reference point in the plane called the **pole** (or origin), and let OX be a fixed horizontal directed ray emanating from O called the **initial line** (or polar axis).

- **Radius Vector (r):** The directed distance from the pole O to any point P , $r = |OP| \geq 0$.
- **Vectorial Angle (θ):** The anti-clockwise angle from the initial line OX to OP , $\theta = \angle XOP$.

The point P is uniquely represented by the ordered pair (r, θ) .

Polar Coordinate System & Projections



1.2 Polar Equation of a Straight Line

Theorem: Standard Polar Equation of a Straight Line

Let p be the length of the perpendicular drawn from the pole O to the straight line RS , and let α be the angle that this normal ON makes with the initial line OX . Then the polar equation of the straight line is:

$$p = r \cos(\theta - \alpha) \iff \frac{p}{r} = \cos(\theta - \alpha) \quad (1)$$

Step-by-Step Proof from Right Triangle $\triangle ONR$

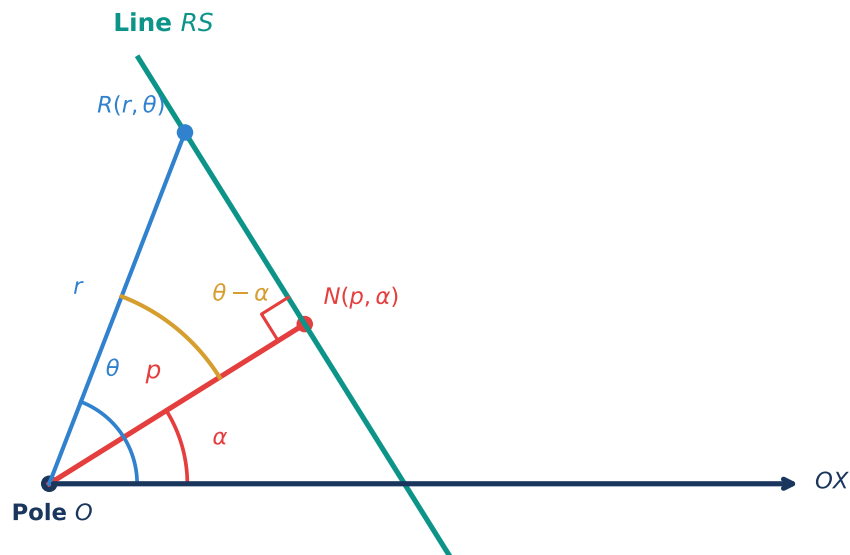
1. Consider a straight line RS . Draw perpendicular $ON \perp RS$ from the pole O .
2. Let $ON = p$, and let the polar coordinates of N be (p, α) , so $\angle XON = \alpha$.
3. Let $R(r, \theta)$ be any point on RS , so $OR = r$ and $\angle XOR = \theta$.
4. In the right-angled triangle $\triangle ONR$:

$$\angle NOR = \angle XOR - \angle XON = \theta - \alpha \quad (2)$$

5. By standard right-triangle trigonometry:

$$\cos(\angle NOR) = \frac{ON}{OR} \implies \cos(\theta - \alpha) = \frac{p}{r} \implies \mathbf{p} = \mathbf{r} \cos(\theta - \alpha) \quad (3)$$

Polar Straight Line: $p = r \cos(\theta - \alpha)$



1.3 Properties and Special Cases of the Polar Straight Line

Important Forms and Special Cases

1. **Cartesian Form:** Expanding $p = r \cos(\theta - \alpha)$:

$$p = r(\cos \theta \cos \alpha + \sin \theta \sin \alpha) \implies \mathbf{p} = \mathbf{x} \cos \alpha + \mathbf{y} \sin \alpha \quad (4)$$

2. **General Linear Form:**

$$\frac{p}{r} = \cos \theta \cos \alpha + \sin \theta \sin \alpha \implies \frac{1}{\mathbf{r}} = \mathbf{A} \cos \theta + \mathbf{B} \sin \theta \quad (5)$$

where $A = \frac{\cos \alpha}{p}$ and $B = \frac{\sin \alpha}{p}$.

3. **Line Perpendicular to Initial Line ($\alpha = 0$):**

$$p = r \cos \theta \iff x = p \quad (6)$$

4. **Line Parallel to Initial Line ($\alpha = \pi/2$):**

$$p = r \cos\left(\theta - \frac{\pi}{2}\right) = r \sin \theta \iff y = p \quad (7)$$

5. **Line Passing Through the Pole ($p = 0$):**

$$\cos(\theta - \alpha) = 0 \implies \theta = \alpha \pm \frac{\pi}{2} = \text{constant} \quad (8)$$

6. **Parallel Lines:** Two parallel lines have the form $r \cos(\theta - \alpha) = p$ and $r \cos(\theta - \alpha) = p'$.

7. **Mutually Perpendicular Lines:** Two perpendicular lines have the form $r \cos(\theta - \alpha) = p$ and $r \cos(\theta - \alpha') = p'$ where $\alpha' \sim \alpha = \pi/2$, i.e., $r \sin(\theta - \alpha) = p'$.

2 Module 2: Polar Equation of a Circle

2.1 General Equation of a Circle

Theorem: General Polar Equation of a Circle

The polar equation of a circle of radius a with center at $C(R, \alpha)$ is:

$$r^2 - 2rR \cos(\theta - \alpha) + R^2 - a^2 = 0 \quad (9)$$

Step-by-Step Proof using the Law of Cosines on $\triangle POC$

1. Let O be the pole, OX the initial line.
2. Let $C(R, \alpha)$ be the center of the circle, so $OC = R$ and $\angle XOC = \alpha$.
3. Let $P(r, \theta)$ be any point on the circle, so $OP = r$, $CP = a$ (radius), and $\angle XOP = \theta$.
4. In triangle $\triangle POC$, the included angle is $\angle POC = \theta - \alpha$.
5. By the Law of Cosines in $\triangle POC$:

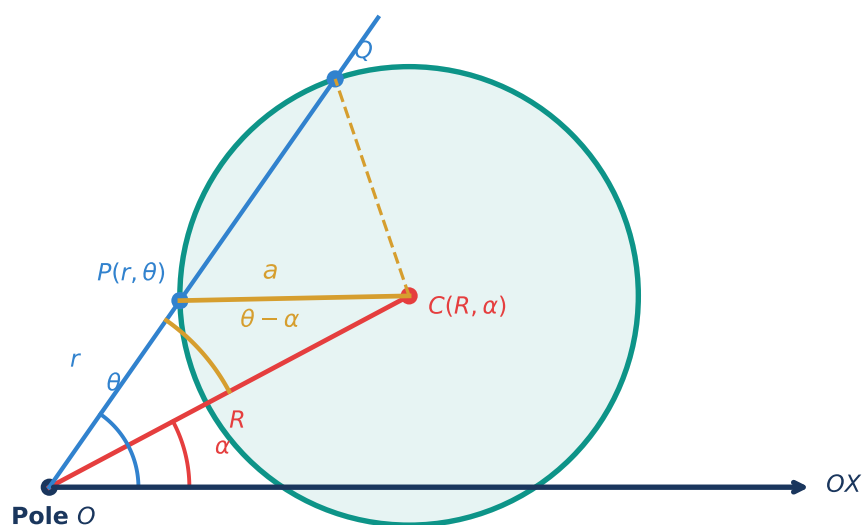
$$CP^2 = OP^2 + OC^2 - 2(OP)(OC) \cos(\angle POC) \quad (10)$$

6. Substituting $CP = a$, $OP = r$, $OC = R$:

$$a^2 = r^2 + R^2 - 2rR \cos(\theta - \alpha) \quad (11)$$

$$r^2 - 2rR \cos(\theta - \alpha) + R^2 - a^2 = 0$$

Polar Circle: $r^2 - 2rR \cos(\theta - \alpha) + R^2 - a^2 = 0$



2.2 Standard Particular Cases of the Polar Circle

Particular Configurations of the Circle

1. Center on Initial Line ($\alpha = 0$):

$$r^2 - 2rR \cos \theta + R^2 - a^2 = 0 \quad (12)$$

2. Pole on Circumference ($R = a$):

$$r^2 - 2ra \cos(\theta - \alpha) = 0 \implies \mathbf{r = 2a \cos(\theta - \alpha)} \quad (13)$$

3. Pole on Circumference and Initial Line is Diameter ($\alpha = 0, R = a$):

$$\mathbf{r = 2a \cos \theta} \quad (14)$$

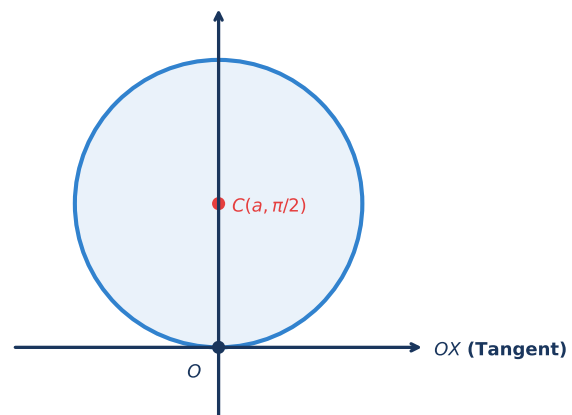
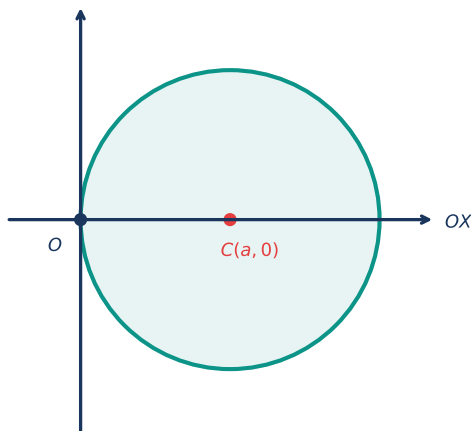
4. Pole on Circumference and Initial Line is Tangent at Pole ($\alpha = \pi/2, R = a$):

$$r = 2a \cos\left(\theta - \frac{\pi}{2}\right) \implies \mathbf{r = 2a \sin \theta} \quad (15)$$

5. Pole at Center ($R = 0$):

$$r^2 - a^2 = 0 \implies \mathbf{r = a} \quad (16)$$

Case 1: $r = 2a \cos \theta$ (Pole on circle, OX along diameter) **Case 2:** $r = 2a \sin \theta$ (Pole on circle, OX is tangent)



3 Module 3: Polar Equation of a Conic (Focus as Pole)

3.1 Standard Conic Equation $\frac{l}{r} = 1 + e \cos \theta$

Theorem: Polar Equation of a Conic

Let the focus S be taken as the pole, and let SX perpendicular to the directrix XM be the initial line. Then the polar equation of the conic is:

$$\frac{l}{r} = 1 + e \cos \theta \quad (17)$$

where e is the eccentricity and l is the semi-latus rectum.

Step-by-Step Proof from Focus-Directrix Property

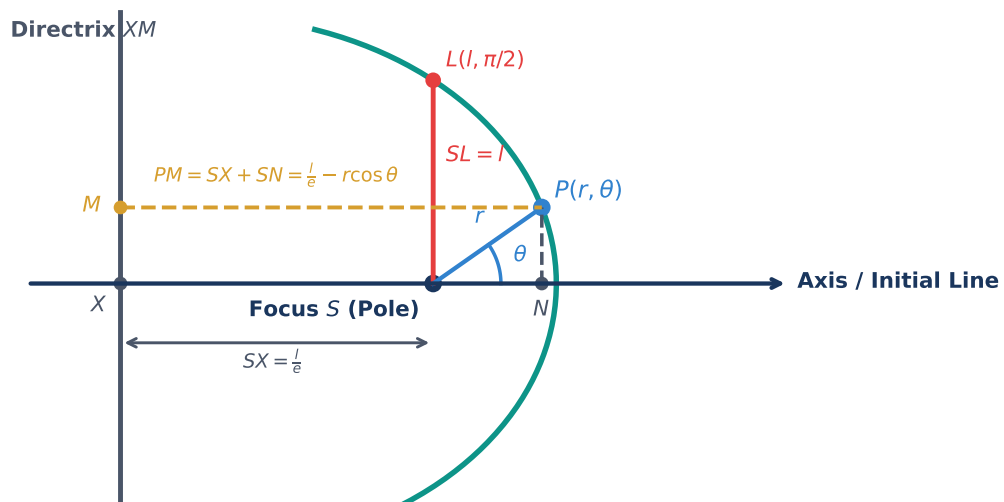
1. Let S be the focus (pole), XM the directrix, and SX the initial line perpendicular to XM .
2. Let $P(r, \theta)$ be any point on the conic. Draw $PN \perp SX$ and $PM \perp XM$.
3. Let $SL = l$ be the semi-latus rectum ($\theta = \pi/2$).
4. By definition of a conic: $\frac{SP}{PM} = e \implies SP = e \cdot PM$.
5. For point L : $\frac{SL}{SX} = e \implies l = e \cdot SX \implies SX = \frac{l}{e}$.
6. From the geometry of projections:

$$PM = NX = SX + SN \quad (18)$$

7. In $\triangle SNP$: $\angle PSN = \pi - \theta$, so $SN = r \cos(\pi - \theta) = -r \cos \theta$.
8. Therefore $PM = \frac{l}{e} - r \cos \theta$.
9. Substituting PM into $r = e \cdot PM$:

$$r = e \left(\frac{l}{e} - r \cos \theta \right) = l - er \cos \theta \implies r(1 + e \cos \theta) = l \implies \frac{l}{r} = 1 + e \cos \theta \quad (19)$$

Conic Focus-Directrix Projection: $\frac{l}{r} = 1 + e \cos \theta$



3.2 Alternative Orientations and Directrices

Orientations and Directrices

1. Axis along Negative Direction:

$$\frac{1}{r} = 1 - e \cos \theta \quad (20)$$

2. Axis Inclined at Angle α to Initial Line:

$$\frac{1}{r} = 1 + e \cos(\theta - \alpha) \quad (21)$$

3. Polar Equation of Directrix (Corresponding to Focus S):

$$\frac{1}{r} = e \cos \theta \quad (22)$$

4. Other Directrix (Central Conics):

$$\frac{1}{r} = -\frac{e(1 - e^2)}{1 + e^2} \cos \theta \quad (23)$$

4 Module 4: Chords, Tangents, Normals, and Chords of Contact

4.1 Equation of the Chord Joining Two Points

Theorem: Equation of Chord of a Conic

The equation of the chord joining points $P(r_1, \alpha - \beta)$ and $Q(r_2, \alpha + \beta)$ on the conic $\frac{l}{r} = 1 + e \cos \theta$ is:

$$\frac{l}{r} = e \cos \theta + \sec \beta \cos(\theta - \alpha) \quad (24)$$

Step-by-Step Derivation of Chord Equation

1. Let the equation of the line PQ be $\frac{l}{r} = A \cos \theta + B \sin \theta$.

2. Since $P(\alpha - \beta)$ and $Q(\alpha + \beta)$ lie on both the conic and line:

$$(A - e) \cos(\alpha - \beta) + B \sin(\alpha - \beta) = 1 \quad \text{--- (1)} \quad (25)$$

$$(A - e) \cos(\alpha + \beta) + B \sin(\alpha + \beta) = 1 \quad \text{--- (2)} \quad (26)$$

3. Subtracting (1) from (2) yields $\frac{A-e}{\cos \alpha} = \frac{B}{\sin \alpha} = k$.

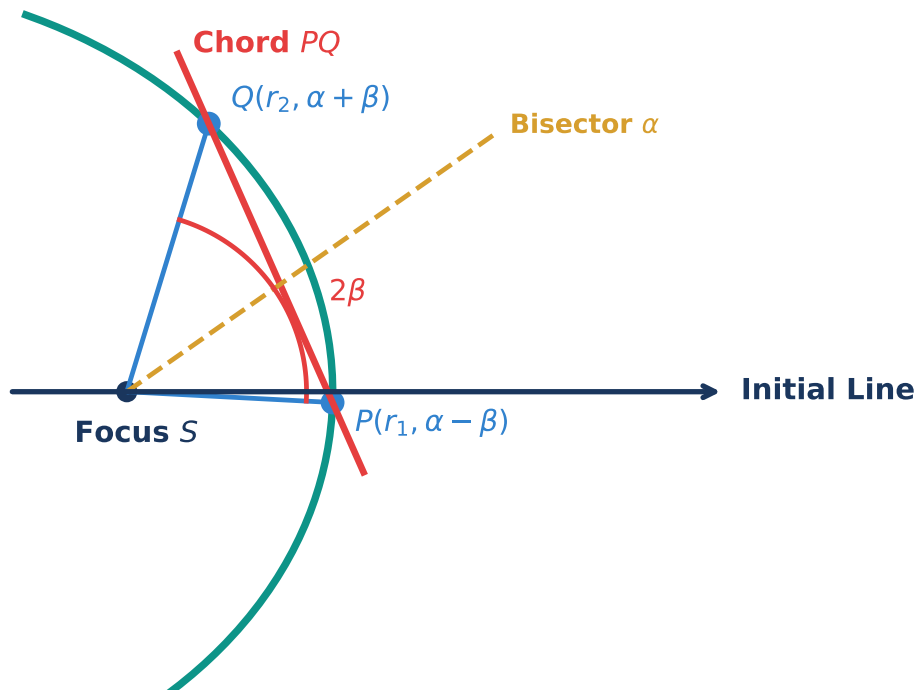
4. Substituting into (1) gives $k \cos \beta = 1 \implies k = \sec \beta$.

5. Hence $A = e + \cos \alpha \sec \beta$ and $B = \sin \alpha \sec \beta$.

6. Substituting back into line equation:

$$\frac{l}{r} = e \cos \theta + \sec \beta \cos(\theta - \alpha) \quad (27)$$

Chord of a Conic: $\frac{l}{r} = e \cos \theta + \sec \beta \cos(\theta - \alpha)$



4.2 Equations of Tangent and Normal

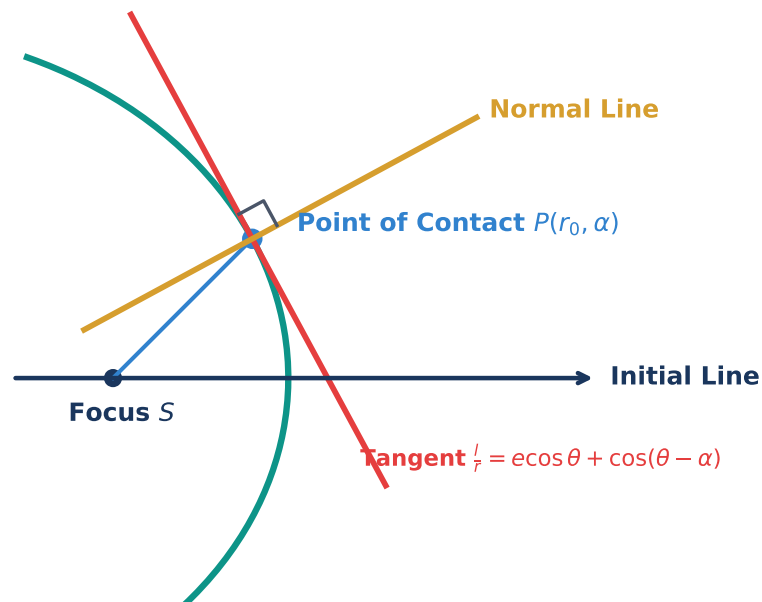
Tangent and Normal to a Conic

1. Tangent at Point
- (α)
- (Limiting position
- $\beta \rightarrow 0$
-):

$$\frac{1}{r} = e \cos \theta + \cos(\theta - \alpha) \quad (28)$$

2. Normal at Point of Contact
- (α)
- :

$$\frac{le \sin \alpha}{1 + e \cos \alpha} \cdot \frac{1}{r} = e \sin \theta + \sin(\theta - \alpha) \quad (29)$$

Tangent and Normal to a Conic at Point of Contact $P(\alpha)$ 

4.3 Chord of Contact of Tangents from an External Point

Theorem: Chord of Contact

The chord of contact of tangents drawn from an external point $A(r_1, \theta_1)$ to the conic $\frac{l}{r} = 1 + e \cos \theta$ is:

$$\left(\frac{1}{r} - e \cos \theta\right) \left(\frac{1}{r_1} - e \cos \theta_1\right) = \cos(\theta - \theta_1) \quad (30)$$

Step-by-Step Proof of Chord of Contact

1. Let tangents from $A(r_1, \theta_1)$ touch the conic at $P(\alpha - \beta)$ and $Q(\alpha + \beta)$.

2. Equations of tangents at P and Q are:

$$\frac{l}{r} = e \cos \theta + \cos(\theta - \alpha + \beta) \quad (31)$$

$$\frac{l}{r} = e \cos \theta + \cos(\theta - \alpha - \beta) \quad (32)$$

3. Since both pass through $A(r_1, \theta_1)$:

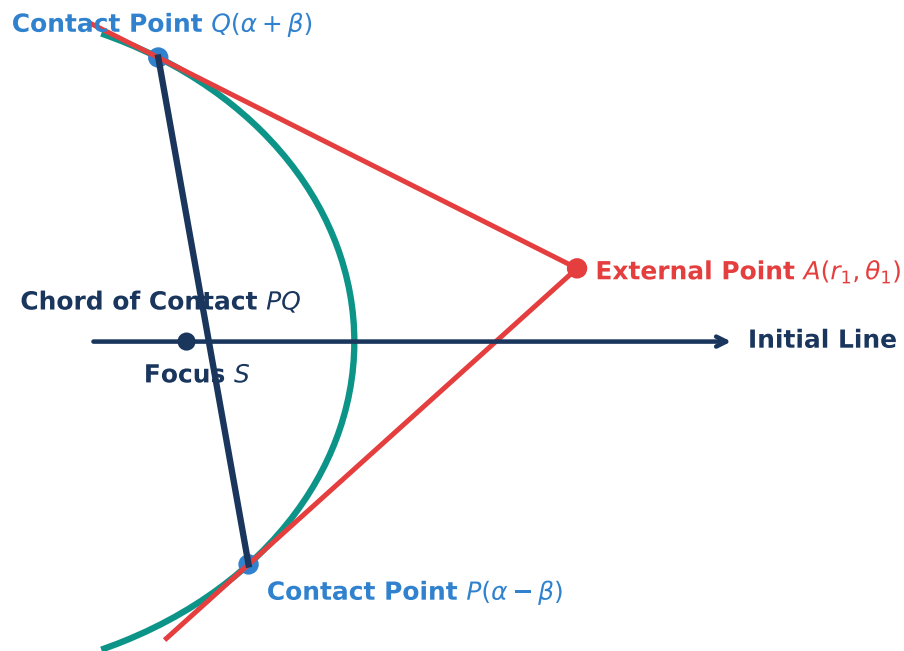
$$\cos(\theta_1 - \alpha + \beta) = \cos(\theta_1 - \alpha - \beta) = \frac{l}{r_1} - e \cos \theta_1 \quad (33)$$

4. This gives $\alpha = \theta_1 - n\pi$ and $\sec \beta = \frac{(-1)^n}{\frac{l}{r_1} - e \cos \theta_1}$.

5. Substituting α and $\sec \beta$ into the chord equation $\frac{l}{r} = e \cos \theta + \sec \beta \cos(\theta - \alpha)$:

$$\left(\frac{1}{r} - e \cos \theta\right) \left(\frac{1}{r_1} - e \cos \theta_1\right) = \cos(\theta - \theta_1) \quad (34)$$

Chord of Contact of Tangents from External Point $A(r_1, \theta_1)$



5 Module 5: Fully Solved Examination Practice Problems

Problem 1: Determining the Nature and Parameters of a Polar Conic

Question: Identify the nature, eccentricity, and semi-latus rectum of the conic given by:

$$\frac{8}{r} = 4 - 5 \cos \theta \quad (35)$$

Solution:

1. Divide the entire equation by 4:

$$\frac{8/4}{r} = \frac{4}{4} - \frac{5}{4} \cos \theta \implies \frac{2}{r} = 1 - \frac{5}{4} \cos \theta \quad (36)$$

2. Comparing with $\frac{l}{r} = 1 - e \cos \theta$:

- Semi-latus rectum: $l = 2$
- Eccentricity: $e = \frac{5}{4} = 1.25$

3. Since $e = \frac{5}{4} > 1$, the conic is a **Hyperbola**.

Problem 2: Finding Points on a Conic with Given Radius Vector

Question: Find the points on the conic $\frac{14}{r} = 3 - 8 \cos \theta$ whose radius vector is $r = 2$.

Solution:

1. Substitute $r = 2$:

$$\begin{aligned} \frac{14}{2} &= 3 - 8 \cos \theta \implies 7 - 3 = -8 \cos \theta \\ 4 &= -8 \cos \theta \implies \cos \theta = -\frac{1}{2} \end{aligned} \quad (37)$$

2. Solving for θ :

$$\cos \theta = \cos \left(\pi - \frac{\pi}{3} \right) = \cos \left(\frac{2\pi}{3} \right) \implies \theta = \pm \frac{2\pi}{3} \quad (38)$$

3. Therefore, the required points on the conic are:

$$\mathbf{P}_1 = \left(2, \frac{2\pi}{3} \right) \quad \text{and} \quad \mathbf{P}_2 = \left(2, -\frac{2\pi}{3} \right) \equiv \left(2, \frac{4\pi}{3} \right) \quad (39)$$

6 Module 6: Master Quick Revision Cheat Sheet

1. Polar Coordinate Foundations and Distance Formulas

- **Cartesian to Polar Relations:**

$$x = r \cos \theta, \quad y = r \sin \theta, \quad r = \sqrt{x^2 + y^2}, \quad \theta = \arctan(y/x)$$

- **Distance Between Two Points $P_1(r_1, \theta_1)$ and $P_2(r_2, \theta_2)$:**

$$d = \sqrt{r_1^2 + r_2^2 - 2r_1r_2 \cos(\theta_1 - \theta_2)}$$

- **Area of Triangle $\triangle OP_1P_2$ with Pole as Vertex:**

$$\Delta = \frac{1}{2}r_1r_2|\sin(\theta_2 - \theta_1)|$$

- **Collinearity of Three Points P_1, P_2, P_3 :**

$$\frac{\sin(\theta_2 - \theta_3)}{r_1} + \frac{\sin(\theta_3 - \theta_1)}{r_2} + \frac{\sin(\theta_1 - \theta_2)}{r_3} = 0$$

2. Straight Lines and Circles in Polar Coordinates

- **Standard Straight Line (Normal Form):**

$$p = r \cos(\theta - \alpha)$$

where p is the length of the normal ON , and α is the inclination angle of ON .

- **General Linear Form:**

$$\frac{1}{r} = A \cos \theta + B \sin \theta \quad \left(A = \frac{\cos \alpha}{p}, B = \frac{\sin \alpha}{p} \right)$$

- **Special Straight Lines:**

- Line through pole ($p = 0$): $\theta = \text{constant}$
- Line perpendicular to initial line ($\alpha = 0$): $r \cos \theta = p \iff x = p$
- Line parallel to initial line ($\alpha = \pi/2$): $r \sin \theta = p \iff y = p$

- **General Polar Circle (Center $C(R, \alpha)$, Radius a):**

$$r^2 - 2rR \cos(\theta - \alpha) + R^2 - a^2 = 0$$

- **Standard Circle Special Cases:**

- Center on initial line ($\alpha = 0$): $r^2 - 2rR \cos \theta + R^2 - a^2 = 0$
- Pole on circumference ($R = a$): $r = 2a \cos(\theta - \alpha)$
- Initial line is diameter: $r = 2a \cos \theta$
- Initial line is tangent at pole: $r = 2a \sin \theta$
- Center at pole ($R = 0$): $r = a$

3. Conic Sections Referred to Focus as Pole

- **Standard Conic Equation:**

$$\frac{l}{r} = 1 + e \cos \theta \quad (\text{Focus at pole, directrix at } x = -l/e)$$

- **Conic Classification by Eccentricity e :**

- $e = 0 \implies$ **Circle**
- $0 < e < 1 \implies$ **Ellipse**
- $e = 1 \implies$ **Parabola**
- $e > 1 \implies$ **Hyperbola**

- **Alternative Conic Orientations:**

- Directrix on positive side: $\frac{l}{r} = 1 - e \cos \theta$
- Major axis inclined at angle α : $\frac{l}{r} = 1 + e \cos(\theta - \alpha)$

- **Directrix Equations:** $\frac{l}{r} = e \cos \theta$ (corresponding directrix) and $\frac{l}{r} = -\frac{e(1-e^2)}{1+e^2} \cos \theta$ (other directrix).

- **Chord Joining Points $P(\alpha - \beta)$ and $Q(\alpha + \beta)$:**

$$\frac{l}{r} = e \cos \theta + \sec \beta \cos(\theta - \alpha)$$

- **Tangent at Point of Contact (α) (Limiting Case $\beta \rightarrow 0$):**

$$\frac{l}{r} = e \cos \theta + \cos(\theta - \alpha)$$

- **Normal at Point of Contact (α) :**

$$\frac{le \sin \alpha}{1 + e \cos \alpha} \cdot \frac{1}{r} = e \sin \theta + \sin(\theta - \alpha)$$

- **Chord of Contact from External Point $A(r_1, \theta_1)$:**

$$\left(\frac{l}{r} - e \cos \theta \right) \left(\frac{l}{r_1} - e \cos \theta_1 \right) = \cos(\theta - \theta_1)$$

- **Polar of Point (r_1, θ_1) :** $\left(\frac{l}{r} - e \cos \theta \right) \left(\frac{l}{r_1} - e \cos \theta_1 \right) = \cos(\theta - \theta_1)$.